COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 07

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Construct a complete truth table to decide whether $(p \to q) \oplus (q \to p)$ and $p \oplus q$ are logically equivalent.

(b) **[2 points]** Are $p \to q$ and its converse always equivalent? Give an assignment showing your answer and also evaluate the inverse at that assignment.

(c) **[5 points]** From $p \wedge q$, $(p \vee r) \to s$, and $q \to t$, derive $s \wedge t$ using named inference rules. Your proof must explicitly use addition.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** Let $D = \{-1, 0, 1\}$ and $P(x)$ mean $x^2 = 1$. Expand $\forall x \in D\, P(x)$ and $\exists x \in D\, P(x)$ as propositional conjunction/disjunction and give their truth values (2 points). Give both truth values when the domain is empty, with justification (2 points).

(b) **[4 points]** Negate $\forall x \exists y \forall z((P(x, y) \vee Q(z)) \wedge R(x, y, z))$, with negations immediately before predicates only.

(c) **[5 points]** Rename the bound variables in $(\forall x P(x)) \vee (\exists x Q(x))$ so the quantifiers use different names (1 point). Then prove that $\forall x(P(x) \rightarrow Q(x))$ and $\exists x \neg Q(x)$ imply $\exists x \neg P(x)$ (4 points).

Problem 3: Sets and functions [12 points]

(a) **[2 points]** For $i = 0, 1, 2$ let $A_i = \{i, i+1\}$. Compute $\bigcup_{i=0}^{2} A_i$ and $\bigcap_{i=0}^{2} A_i$.

(b) **[3 points]** If $f : A \to B$ and $g : B \to C$ are injective, prove $g \circ f$ is injective (2 points). Does injectivity of $g \circ f$ alone force g to be injective? Give a counterexample if not (1 point).

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ for all sets A, B.

(ii) $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

(iii) The set of finite strings over a finite nonempty alphabet is countable.

(iv) The set of infinite binary sequences with a fixed finite number of ones is countable.

(v) $\mathbb{N}^3$ is uncountable.

(vi) If $g \circ f$ is injective, then f is injective.

(vii) A function $A \to B$ may assign two different outputs in B to one input in A.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** A visitor tours six distinct cities exactly once, beginning at a specified city A and ending at a different specified city B. Count the visit orders.

(b) **[2 points]** Count functions from a labelled four-element set to a labelled five-element set whose range has exactly three elements.

(c) **[2 points]** Count length-eight bit strings with at most one 1.

(d) **[2 points]** Arrange four identical red tokens and three identical blue tokens in a line, with red first and blue last. Count the colour strings.

(e) **[2 points]** Seven distinct people sit around a circle, and A, B, C must occur consecutively in that clockwise order. Rotations are equivalent; reflections are different. Count the seatings.

(f) **[2 points]** Count nonnegative integer triples (x, y, z) satisfying $x + y + z \leq 6$.

(g) **[2 points]** Find the constant term in $(x^3 + 2x^{-2})^{10}$.

(h) **[2 points]** Eight distinct files enter labelled folders A, B of sizes three and two, leaving three outside. Exactly one of the specified files X, Y must be outside. Within-folder order does not matter. Count the assignments.

Problem 5: Combinatorial proof [10 points]

(a) **[6 points]** For $1 \leq r < n$, prove Pascal's identity by a disjoint partition of subsets:

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}.$$

(b) **[4 points]** Let $p(a, b)$ count right/up unit-step paths from $(0, 0)$ to (a, b). Explain its Pascal recurrence and boundary values, then count paths to $(4, 3)$ avoiding $(2, 1)$.

Problem 6: Inclusion-exclusion [9 points]

[9 points] Strings have six positions, with symbols from $\{A, B, C, D\}$; repetition is allowed. Use inclusion–exclusion to count strings whose endpoints differ and in which each of A, B, C appears at least once.

Problem 7: Pigeonhole principle [7 points]

[7 points] Among any 65 distinct bit strings of length seven, prove that two agree in their first six positions and differ only in the last position. Show that 64 strings do not suffice to force this.

Problem 8: Unions of divisibility events [6 points]

Choose an integer uniformly from $S = \{1, \ldots, 60\}$. Let A, B, C be divisibility by $2, 3, 5$, respectively.

(a) **[1 points]** Find $P(A), P(B), P(C)$.

(b) **[1 points]** Find $P(A \cap B)$ and test independence of A, B.

(c) **[1 points]** Find $P(A \cap C)$ and $P(B \cap C)$.

(d) **[1 points]** Find $P(A \cap B \cap C)$.

(e) **[2 points]** Use inclusion–exclusion to find the probability of divisibility by at least one of $2, 3, 5$.

Problem 9: Nonuniform prize locations [7 points]

A prize is behind doors $1, 2, 3$ with probabilities $1/2, 1/3, 1/6$. You pick door 1. The host always opens an unchosen goat door, uniformly if both are goats and otherwise the unique legal door. Let M denote the opening.

(a) **[3 points]** Given $M = 2$, find the stay/switch posteriors and choose the better action.

(b) **[4 points]** Choose the better action given $M = 3$, and find the overall success probability of the policy using both optimal actions.

Problem 10: Pairwise independence suffices [10 points]

Independent fair bits U, V give four equiprobable outcomes. Define $A = U$, $B = V$, and $C = 1$ if $U \neq V$, otherwise 0. These are numerical indicators.

(a) **[2 points]** Give the distribution, mean and variance of C.

(b) **[2 points]** Check independence of A, B using all joint-value probabilities.

(c) **[2 points]** Check that A, C and B, C are independent. Are all three mutually independent?

(d) **[4 points]** Find $E(A+B-C)$ and $V(A+B-C)$. Explain why the weaker independence established above is sufficient for the variance calculation.

Draft Paper

Mock examination working space.

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